Dr. Luis J. Gilarranz

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Birthplace: July 9th, 1984. Segovia (Spain)

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Education

2010 - Nov 2015 **PhD, Ecology**, Universidad de Alcalá de Henares. Spain. Advisor: Prof. Dr. Jordi Bascompte.

2009 - 2010 **MSc, Physics of Complex Systems**. Universidad Politécnica de Madrid. Spain. Advisor: Prof. Dr. Javier Galeano

2004 - 2009 **BSc + MSc, Environmental Sciences.** Universidad de Alcalá de Henares. Spain.

Academic positions and Research appointments

2017 - **Postdoctoral researcher,** Independent Position. Eawag, Dübendorf, Switzerland.

2016 - 2017 **Postdoctoral researcher,** Prof. Dr. Jordi Bascompte. University of Zurich, Zürich, Switzerland.

Nov 2014 **Visiting scientist,** Prof. Dr. George Sugihara. University of California at San Diego. San Diego, USA.

Feb - Apr 2013 **Visiting scientist,** Prof. Dr. Andrew Gonzalez. McGill University. Montreal, Canada.

Oct - Nov 2011 **Visiting scientist,** Prof. Dr. Mathew Leibold. University of Texas. Austin, USA.

Publications (five most significant are marked in blue)

14. B. Baiser, D. Gravel, A. Cirtwill, J. Dunne, A. Fahimipour, **L.J. Gilarranz**, J. Grochow, N. Martinez, T. Poisot, T. Romanuk, D. Stouffer, F. Valdovinos, R. Williams, S. Wood, J. Yeakel. The Macroecology of Food Webs.
Submitted.

13. B. Bramon-Mora, D. Gravel, **L.J. Gilarranz**, T. Poisot, D. Stouffer. Identifying a common backbone of interactions underlying food webs from different ecosystems.
Nature Communications. *In press.*

12. **L.J. Gilarranz**, B. Rayfield, G. Liñán-Cembrano, J. Bascompte, and A. Gonzalez. Network modularity buffers the spread of perturbations in metapopulations.
Science *357: 199-201 (2017)*

We experimentally demonstrate how certain network structures confine perturbations within few nodes. However, time-discrete models show that in the absence of perturbations these network structures are detrimental to system's performance.

11. J.E. Lavabre, **L.J. Gilarranz**, M.A. Fortuna, and J. Bascompte. The functional diversity of frugivorous birds shapes the spatial pattern of seed dispersal in a relict plant species.
Philosophical Transactions of the Royal Society B. 371: 20150280 (2016)

10. **L.J. Gilarranz**, C. Mora, and J. Bascompte. Anthropogenic effects are associated to a lower persistence of marine food webs.
Nature Communications, 7, 10737 (2016)

We show how human-related impacts like pollution, fishing, or climate change, diminishes the persistence of marine food webs by altering species biomass and food web structure. To show this we combine dynamic models, complex networks, GIS, and mixed effects statistical models.

9. M.A. Aizen, G. Gleiser, M. Sabatino, **L.J. Gilarranz**, J. Bascompte, and M. Verdú. The phylogenetic structure of plant-pollinator networks increases with habitat size and isolation.
Ecology Letters, 19: 29-36 (2016)

We show how dispersal patterns that follow metrics extracted from complex networks explain the phylogenetic structure of plant-pollinator networks in fragmented landscapes.

8. H. Ye, E.R. Deyle, **L.J. Gilarranz**, and G. Sugihara. Distinguishing time-delayed causal interactions using convergent cross mapping.
Scientific Reports, 5, 14750 (2015)

We developed a new method to determine whether there is a causal relationship between two variables just from observational data (time series). It also allows us to determine the time that takes an event in one variable to be reflected by the time series of the other variable.

7. **L.J. Gilarranz**, M. Sabatino, M.A. Aizen, and J. Bascompte. Hotspots of Mutualistic Networks.
Journal of Animal Ecology, 84, 407–413 (2015)

Networks of networks. Our theoretical results match what is found in nature. Number of species, number of interactions, and the local network structure are more related with the betweenness centrality of the habitat fragment than with any other variable of habitat size or isolation.

6. **L.J. Gilarranz**, A. Hastings, and J. Bascompte. Inferring topology from dynamics in spatial networks.
Theoretical Ecology, 8:15–21 (2015)

5. S. Saavedra, **L.J. Gilarranz**, R.P. Rohr, M. Schnabel, B. Uzzi, and J. Bascompte. Stock fluctuations are correlated and amplified across networks of interlocking directorates.
EPJ Data Science, 3:30 (2014)

4. S. Saavedra, R.P. Rohr, **L.J. Gilarranz**, and J. Bascompte. How structurally stable are global socioeconomic systems?
Journal of the Royal Society Interface, 11:20140693 (2014)

3. **L.J. Gilarranz**, J.J. Lever, R.P. Rohr, and M.A. Fortuna. NextGen VOICES: How will the practice of science change in your lifetime? What will improve and what new challenges will emerge?
Science, 335:37 (2012)

2. **L.J. Gilarranz** and J. Bascompte. Spatial network structure and metapopulation persistence.
Journal of Theoretical Biology, 297:11-16 (2012)

1. **L.J. Gilarranz**, J. Galeano, and J.M. Pastor. The architecture of mutualistic weighted networks.
Oikos, 121:1154–1162 (2012)

Awarded fellowships & projects (Total amount acquired: 602.500 CHF)

Sep 2017-	Eawag. Eawag Postdoctoral Research Fellowship.	200.000 CHF
Oct 2010 - Aug 2014	Spanish Ministry of Science and Innovation. FPU.	70.000 CHF
Jul 2010 - Sep 2010	Spanish National Research Council. JAE-Predoc.	70.000 CHF
Oct 2008 - Nov 2008	Spanish National Research Council. JAE-Intro.	2.500 CHF
2017	European Commission. Marie-Sklodowska-Curie Individual Fellowship. Awarded but declined due to incompatibility with my current position	205.000 CHF
2017	Spanish Ministry of Economy. Juan de la Cierva. Awarded but declined due to incompatibility with my current position	55.000 CHF

Invited talks and lectures

2017 Institut des Sciences de l'Evolution. CNRS. Montpellier, France
2017 Institute for Chemistry and Biology of the Marine Environment. Carl von Ossietzky University Oldenburg. Germany
2017 Department of Civil Engineering, Massachusetts Institute of Technology. Cambridge, USA
2017 Department of Evolutionary Biology and Environmental Studies. University of Zurich. Zurich, Switzerland
2017 Swiss Federal Institute of Aquatic Science and Technology. Kastanienbaum, Switzerland
2015 NetSci 2015 Satellite. Complex Networks in Ecology. Zaragoza, Spain
2015 Workshop on Gradient-Based Ecological Network Research. Santa Fe Institute, New Mexico, USA
2014 Workshop on Networks on networks. University of Göttingen, Germany
2014 Workshop on Network modeling. Stockholm Resilience Center, Sweden
2013 Ecological Society of America Annual Meeting. Minneapolis, Minnesota, USA
2013 Luis Santaló school: Mathematics of Planet Earth. UIMP, Santander, Spain
2011 Estación Biológica de Doñana, CSIC. Sevilla, Spain
2011 Section of Integrative Biology. University of Texas, Austin, USA
2009 Centro de Investigación Forestal, INIA. Madrid, Spain

Contributed talks

2017 Association for the Sciences of Limnology and Oceanography. Honolulu, Hawaii, USA
2017 Biology 17. Bern, Switzerland
2015 Association for the Sciences of Limnology and Oceanography. Granada, Spain
2014 Ecological Society of America Annual Meeting. Sacramento, California, USA
2013 European Conference on Complex Systems. Barcelona, Spain
2013 Ecological Society of America Annual Meeting. Minneapolis, Minnesota, USA
2012 Ecological Society of America Annual Meeting. Portland, Oregon, USA
2011 European Ecological Federation Congress. Ávila, Spain
2011 Ecological Society of America Annual Meeting. Austin, Texas, USA
2010 Ecological Society of America Annual Meeting. Pittsburgh, Pennsylvania, USA

Teaching

2018 Teaching a doctorate course in the University of Granada (Spain) about "Complex adaptive systems at multiple scales in evolutionary and global ecology,"

Co-teaching a course for bachelor students in the University of Zurich about "Ecological networks."

2016 Co-teaching two courses in the master program in Environmental Sciences at the University of Zurich: "classic papers in ecology and evolution", and "theoretical ecology".

2013 Invited professor at the Mathematics of Planet Earth course, Organized by the Universidad Internacional Menendez Pelayo, the Luis Santaló School from the Spanish Royal Mathematical Society, and the Physics and Mathematics department from the University of Granada.

Outreach talks and publications

2011 Pecha Kucha Night Sevilla, vol. X. Explaining my research to the general public is equally challenging and gratifying. Pecha Kucha is defined as the art of concise presentations. 20 slides and 20 seconds per slide, a full theater with over 500 attendees, and sharing the stage with 9 other speakers from the most diverse backgrounds.

2014 Un tránsito de lo muy local a lo internacional. Líneas de investigación en la Estación Biológica de Doñana. *Quercus*, 343:52-60

Experience as reviewer

During the last five years I have reviewed manuscripts for the following journals: Nature Communications, Ecology Letters, PLOS Computational Biology, Journal of Animal Ecology, Methods in Ecology and Evolution, PlosONE, Ecology, Ecography, Biological Conservation, Landscape Ecology, Oecología, Aquatic Ecology, Environmental Modelling and Software, Journal of Complex Networks, Journal of Theoretical Biology, The Journal of Network Theory in Finance, Theoretical Ecology, and Advances in Ecological Research.

References (in alphabetical order)

Prof. Dr. Florian Altermatt (Academic mentor): florian.altermatt@eawag.ch
Department of Aquatic Ecology. EAWAG. Überlandstrasse 133, 8600 Dübendorf. Switzerland

Prof. Dr. Jordi Bascompte (PhD supervisor): jordi.bascompte@ieu.uzh.ch
Department of Evolutionary Biology and Environmental Studies. University of Zurich. Winterthurerstrasse 190, 8057 Zurich. Switzerland

Prof. Dr. Andrew Gonzalez (Academic mentor): andrew.gonzalez@mcgill.ca
Department of Biology. 1205 Docteur Penfield. Montreal. Quebec. Canada

Prof. Dr. George Sugihara (Academic mentor): gsugihara@ucsd.edu
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